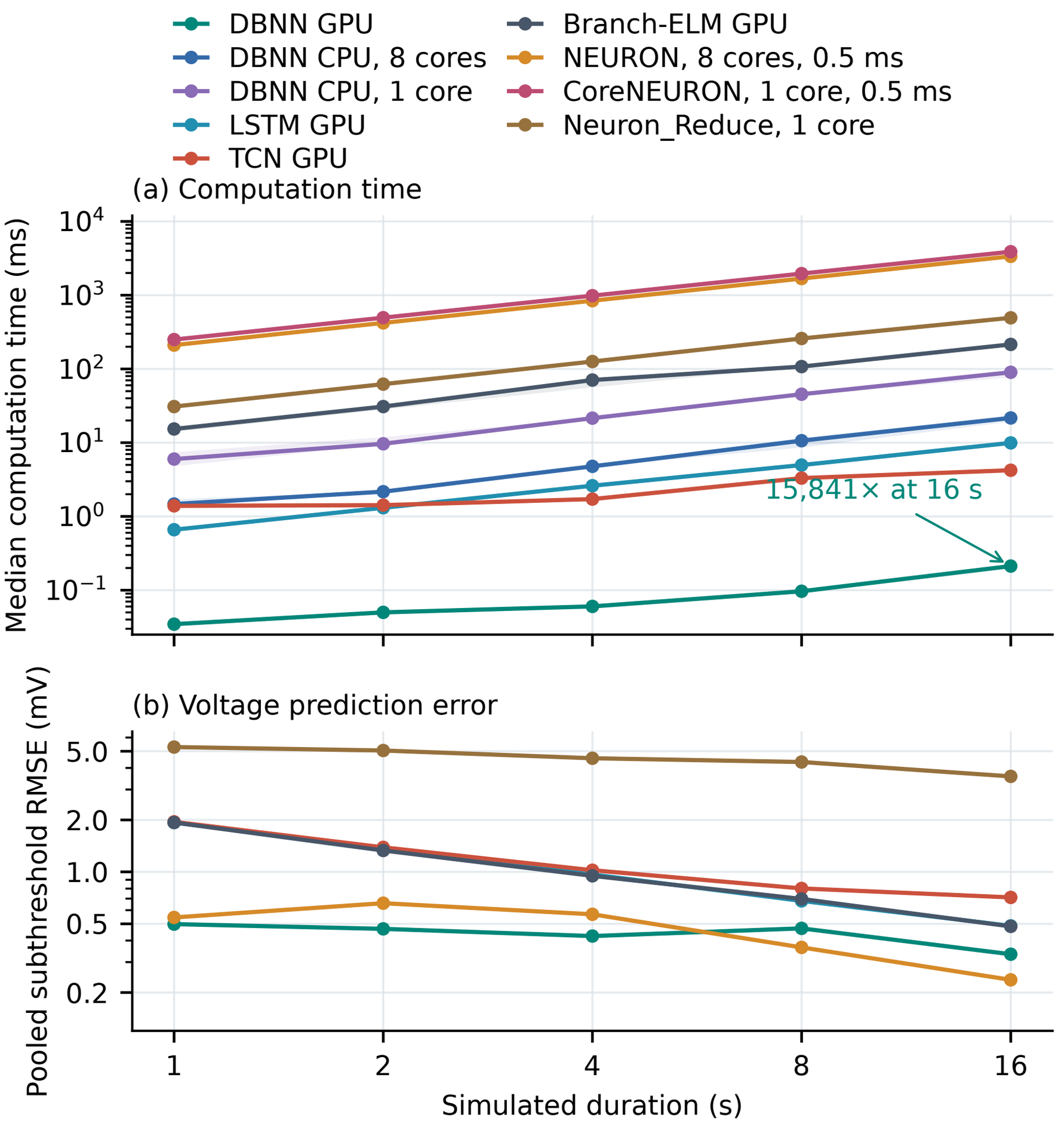
DBNN GPU
DBNN CPU, 8 cores
DBNN CPU, 1 core
LSTM GPU
TCN GPU
Branch-ELM GPU
NEURON, 8 cores, 0.5 ms
CoreNEURON, 1 core, 0.5 ms
Neuron_Reduce, 1 core
(a) Computation time
Median computation time (ms)
10^4
10^3
10^2
10^1
10^0
10^{-1}
15,841× at 16 s
(b) Voltage prediction error
Pooled subthreshold RMSE (mV)
5.0
2.0
1.0
0.5
0.2
1
2
4
8
16
Simulated duration (s)